

A LAW OF THE LOGARITHM FOR NEAREST NEIGHBOUR KERNEL ESTIMATORS IN THE PRESENCE OF CENSORING

Jiang Lu

Department of Statistics and Actuarial Science, University of the Witwatersrand, Private Bag 3, WITS 2050, Johannesburg, South Africa

and

Brenda MacGibbon

Département de Mathématiques, Université du Québec à Montréal, C.P. 8888, Succ. "Centre-ville", Montréal, Quebec H3C 3P8, Canada

Key words: Censoring; density estimate; hazard rate estimate; kernel; law of the logarithm; nearest neighbour

Summary: In this paper we derive a law of the logarithm establishing the exact rate of uniform convergence for nearest neighbour kernel estimators of the density and hazard rate functions in the presence of censoring.

1. Introduction

Let (X_i, Y_i) , $i = 1, 2, \dots, n$, be a sequence of i.i.d. random vectors, where X_i and Y_i are mutually independent with distribution functions F and G and density functions f and g respectively, so that the observations available are the vectors (Z_i, δ_i) , where $Z_i = \min\{X_i, Y_i\}$, $\delta_i = I[X_i \leq Y_i]$, $1 \leq i \leq n$, where $I[A]$ is the indicator function of the set A . Denote $\sum_{i=1}^n I[x \leq Z_i]$ by $\#_n(x)$. For any distribution function F ,

define $\bar{F}(x) = 1 - F(x)$. The well-known Kaplan-Meier (KM) estimate (Kaplan & Meier, 1958) is defined by

$$1 - F_n(x) = \prod_{i: \delta_i=1, Z_i \leq x}^n \left(1 - \frac{1}{\#_n(Z_i)}\right).$$

Our estimate for f will be a kernel-type estimate of the form

$$f_n(x) = \frac{1}{b_n(x)} \int K\left(\frac{x-s}{b_n(x)}\right) dF_n(s), \quad (1)$$

where $b_n(x)$ is a positive bandwidth and K is a smooth probability kernel.

In survival analysis, the hazard rate $\lambda(x) = f(x)/\bar{F}(x)$ is often of primary importance. A problem of considerable interest, therefore, is the estimation of $\lambda(x)$ from a sample of n i.i.d. (possibly censored) lifetimes. The quantity $\Lambda(x) = \int_0^x \lambda(s) ds$ is called the cumulative hazard function. In the censored case, a familiar estimate $\Lambda_n(x)$ of $\Lambda(x)$ is given by

$$\Lambda_n(x) = \sum_{i=1}^n \frac{I[Z_i \leq x, \delta_i = 1]}{\#_n(Z_i)}.$$

For the hazard rate function, just as in (1), a kernel estimator can be defined by the convolution between Λ_n and the kernel function K , that is,

$$\lambda_n(x) = \frac{1}{b_n(x)} \int K\left(\frac{x-s}{b_n(x)}\right) d\Lambda_n(s). \quad (2)$$

Although in most applications $b_n(x)$ has been taken as a constant bandwidth h_n (see Parzen, 1962), improvements of the resulting density and hazard rate estimators can be obtained by varying the bandwidth locally (see, for example, the k_n th nearest neighbour (hereinafter referred to as k -NN) estimator introduced by Loftsgaarden and Quesenberry, 1965). In fact, intuitively, it is obvious that the Parzen method of estimation cannot respond adequately to local variations in $f(x)$; that is, the bandwidth is not data-

responsive and the estimate is unable to adjust to sparseness of data near x since the bandwidth h_n is a constant for every point x . If the bandwidth used is very large, modes of the underlying density will be underestimated and thus detail obscured. If the chosen bandwidth is very small, spurious detail will appear in the resulting density estimate, particularly in the tails. An appropriate choice of bandwidth, between these two extremes is therefore important. In general, better estimates are produced if bandwidths are used which vary between the observations. If a large bandwidth is used at points which occur in the tails of the distribution then the problem of noise in the tails should be overcome, and if a smaller bandwidth can be used at points in the main body of the distribution then the problem of flattening of modes of the distribution should also be alleviated. For an application of kernel estimates with the bandwidths which vary between the observations in calculating centile curves, see Rossiter (1991).

The idea, introduced for censored data by Tanner (1983), is to replace $b_n(x)$ with $a_n(x)$ in (1) and (2), where $a_n(x)$ is the minimum s such that the interval $[x - s, x + s]$ contains k_n uncensored observations from n samples, a given sequence of positive integers. Intuitively, since $a_n(x)$ is large when $f(x)$ is small and vice versa, the k -NN kernel estimator seems to have the desirable property of being self-adjusting to local data fluctuations.

A desired property of estimators is consistency. Referring to $f_n(x)$ in (1) with $b_n(x)$ replaced by the constant bandwidth h_n , the exact rate of pointwise convergence (law of the iterated logarithm) has been derived. For the case of complete data, see Hall (1981) and Stute (1982). For the censored case, see Zhang (1996). On the exact rate of uniform convergence (law of the logarithm) with censored data, Diehl and Stute (1988) established the following

convergence

$$\left(\frac{nh_n}{2 \log h_n^{-1}} \right)^{\frac{1}{2}} \sup_{T'' \leq x \leq T} \sqrt{\frac{\bar{G}(x)}{f(x)}} |f_n(x) - \hat{f}_n(x)| \rightarrow \left(\int K^2(s) ds \right)^{\frac{1}{2}} \text{ a.s.} \quad (3)$$

with $0 < m < f(x) < M$ on a bounded interval $[T'', T']$ with some T' such that $0 \leq T'' < T < T'$, where

$$\hat{f}_n(x) = \frac{1}{h_n} \int K\left(\frac{x-s}{h_n}\right) dF(s).$$

Mack (1983) demonstrated that the analogous result to (3) holds for the k-NN estimate given by

$$f_n(x) = k_n / 2na_n(x),$$

which amounts to using a uniform kernel on $[-1, 1]$ and the bandwidth $a_n(x)$ in (1). Mack's (1983) estimate is for uncensored data. Ralescu (1995) derived the law of the iterated logarithm for the pointwise multivariate nearest neighbour density estimates with complete and censored data. In this paper we extend the original results of Diehl and Stute (1988) and Ralescu (1995); that is, we establish, for a general kernel $K(x)$, the exact rate of uniform convergence with censored data for the k-NN hazard rate estimator as in (2) with $b_n(x)$ replaced by $a_n(x)$ and the k-NN kernel density estimator as in (1) with $b_n(x)$ replaced by $a_n(x)$ respectively. As a result of our methods of proof, we can see, under certain conditions, that kernel estimates with constant bandwidths and k-NN kernel estimates are asymptotically compatible.

Since censored data traditionally occur in lifetime analysis, we do assume that X and Y are nonnegative, though this in no way limits the method. The actually observed Z 's have a distribution function H satisfying $\bar{H}(x) = \bar{F}(x)\bar{G}(x)$. In the following c will represent a generic constant. Note that

$H(T) < 1$, and consider the estimation of f and λ on $[0, T]$. Also in the following, a function g is said to satisfy the Lipschitz condition of order one on an interval $[a, b]$, if there exists a constant $c > 0$, such that for any $x, y \in [a, b]$

$$|g(x) - g(y)| \leq c|x - y|.$$

Define

$$\bar{\lambda}_n(x) = \frac{1}{a_n(x)} \int K\left(\frac{x-s}{a_n(x)}\right) d\Lambda(s),$$

and

$$\bar{f}_n(x) = \frac{1}{a_n(x)} \int K\left(\frac{x-s}{a_n(x)}\right) dF(s).$$

THEOREM 1. Assume that K is a continuously differentiable probability kernel vanishing outside the interval $[-1, 1]$; f and g are bounded on $[0, T']$ for some $T' > T$ and $\bar{H}(T') > 0$; and on $[T'', T']$ where $0 \leq T'' < T < T'$, we have $f \geq m > 0$ and f satisfies a Lipschitz condition of order one on $[T'', T']$. Let k_n satisfy

$$\frac{\log^3 n}{k_n} \rightarrow 0 \text{ and } \frac{\log(n/k_n)}{\log n} \rightarrow c > 0. \quad (4)$$

Then

$$\left(\frac{k_n}{2 \log(n/k_n)}\right)^{\frac{1}{2}} \sup_{T'' \leq x \leq T} \frac{|f_n(x) - \bar{f}_n(x)|}{f(x)} \rightarrow \left(2 \int K^2(s) ds\right)^{\frac{1}{2}} \text{ a.s.} \quad (5)$$

and

$$\left(\frac{k_n}{2 \log(n/k_n)}\right)^{\frac{1}{2}} \sup_{T'' \leq x \leq T} \frac{|\lambda_n(x) - \bar{\lambda}_n(x)|}{\lambda(x)} \rightarrow \left(2 \int K^2(s) ds\right)^{\frac{1}{2}} \text{ a.s.} \quad (6)$$

COROLLARY 1. Under the conditions of theorem 1, if, in addition,

$$k_n^{3/2}/n \log^{\frac{1}{2}}(n/k_n) \rightarrow 0, \quad (7)$$

then

$$\left(\frac{k_n}{2 \log(n/k_n)} \right)^{\frac{1}{2}} \sup_{T'' \leq x \leq T} \frac{|f_n(x) - f(x)|}{f(x)} \rightarrow \left(2 \int K^2(s) ds \right)^{\frac{1}{2}} \text{ a.s.}$$

and

$$\left(\frac{k_n}{2 \log(n/k_n)} \right)^{\frac{1}{2}} \sup_{T'' \leq x \leq T} \frac{|\lambda_n(x) - \lambda(x)|}{\lambda(x)} \rightarrow \left(2 \int K^2(s) ds \right)^{\frac{1}{2}} \text{ a.s.}$$

In the censored case, Ralescu (1995) considered the following k -NN kernel estimate $f_{n1}(x)$ of $f(x)$,

$$f_{n1}(x) = \frac{1}{\rho_n(x)} \int K \left(\frac{x-s}{\rho_n(x)} \right) dF_n(s),$$

where $\rho_n(x)$ is the distance from x to its k_n th nearest neighbour among $Z_1, \dots, Z_n$ instead of the nearest uncensored neighbour as in our case. Note that the difference between $f_{n1}(x)$ and $f_n(x)$ occurs in the definition of their bandwidths $\rho_n(x)$ and $a_n(x)$. We can now define

$$\lambda_{n1}(x) = \frac{1}{\rho_n(x)} \int K \left(\frac{x-s}{\rho_n(x)} \right) d\Lambda_n(s),$$

$$\bar{f}_{n1}(x) = \frac{1}{\rho_n(x)} \int K \left(\frac{x-s}{\rho_n(x)} \right) dF(s),$$

and

$$\bar{\lambda}_{n1}(x) = \frac{1}{\rho_n(x)} \int K \left(\frac{x-s}{\rho_n(x)} \right) d\Lambda(s).$$

Analogous to theorem 1 and corollary 1, we now have the following extensions of Ralescu's univariate results.

THEOREM 2. Under the conditions of theorem 1,

$$\left(\frac{k_n}{2 \log(n/k_n)} \right)^{\frac{1}{2}} \sup_{T'' \leq x \leq T} \frac{|f_{n1}(x) - \bar{f}_{n1}(x)|}{f(x)} \rightarrow \left(2 \int K^2(s) ds \right)^{\frac{1}{2}} \text{ a.s.,} \quad (8)$$

and

$$\left(\frac{k_n}{2 \log(n/k_n)} \right)^{\frac{1}{2}} \sup_{T'' \leq x \leq T} \frac{|\lambda_{n1}(x) - \bar{\lambda}_{n1}(x)|}{\lambda(x)} \rightarrow \left(2 \int K^2(s) ds \right)^{\frac{1}{2}} \text{ a.s.} \quad (9)$$

COROLLARY 2. Under the conditions of corollary 1,

$$\left(\frac{k_n}{2 \log(n/k_n)} \right)^{\frac{1}{2}} \sup_{T'' \leq x \leq T} \frac{|f_{n1}(x) - f(x)|}{f(x)} \rightarrow \left(2 \int K^2(s) ds \right)^{\frac{1}{2}} \text{ a.s.,}$$

and

$$\left(\frac{k_n}{2 \log(n/k_n)} \right)^{\frac{1}{2}} \sup_{T'' \leq x \leq T} \frac{|\lambda_{n1}(x) - \lambda(x)|}{\lambda(x)} \rightarrow \left(2 \int K^2(s) ds \right)^{\frac{1}{2}} \text{ a.s.}$$

2. Proofs

Define

$$\begin{aligned} \alpha_n(x) &= k_n^{\frac{1}{2}} (F_n(x) - F(x)) , \\ h_n(x) &= k_n / (2nf(x)\bar{G}(x)) , \\ f_n^*(x) &= \frac{1}{h_n(x)} \int K \left(\frac{x-s}{h_n(x)} \right) dF_n(s) , \\ \bar{f}_n^*(x) &= \frac{1}{h_n(x)} \int K \left(\frac{x-s}{h_n(x)} \right) dF(s) . \end{aligned}$$

LEMMA 1. Let U be a constant such that $\bar{H}(U) = c > 0$. Let $p_n \rightarrow 0$ and $np_n/\log n \rightarrow \infty$. Then

$$\sup_{\Lambda(A) \leq p_n} |\Lambda_n(A) - \Lambda(A)| = O\left(\sqrt{p_n \log(n)/n}\right) \text{ a.s.,}$$

and

$$\sup_{F(A) \leq p_n} |F_n(A) - F(A)| = O\left(\sqrt{p_n \log(n)/n}\right) \text{ a.s.}$$

The sup is taken over all intervals $A \subset [0, U]$ with mass less than or equal to p_n .

PROOF. See Schäfer (1986).

PROOF OF THEOREM 1. We divide the proof into two steps.

1. First, we prove that

$$(k_n)^{\frac{1}{2}} \sup_{T'' \leq x \leq T} |(f_n(x) - \bar{f}_n(x)) - (f_n^*(x) - \bar{f}_n^*(x))| \rightarrow 0 \text{ a.s.} \quad (10)$$

It follows from the definitions of $f_n(x)$, $\bar{f}_n(x)$, $f_n^*(x)$ and $\bar{f}_n^*(x)$ that the left-hand side of (10) is bounded by

$$\begin{aligned} & \sup_{x \in [T'', T]} \left| \frac{1}{a_n(x)} \int K\left(\frac{x-s}{a_n(x)}\right) d\alpha_n(s) - \frac{1}{h_n(x)} \int K\left(\frac{x-s}{h_n(x)}\right) d\alpha_n(s) \right| \\ & \leq \sup_{x \in [T'', T]} \frac{1}{a_n(x)} \left| \int \left(K\left(\frac{x-s}{a_n(x)}\right) - K\left(\frac{x-s}{h_n(x)}\right) \right) d\alpha_n(s) \right| \\ & \quad + \sup_{x \in [T'', T]} \left| \left(\frac{1}{a_n(x)} - \frac{1}{h_n(x)} \right) \int K\left(\frac{x-s}{h_n(x)}\right) d\alpha_n(s) \right| \\ & = I_1 + I_2. \end{aligned}$$

From Schäfer (1986) and the definitions of $a_n(x)$ and $h_n(x)$, for n large

enough, we have

$$\sup_{x \in [T'', T]} \left| \frac{1}{a_n(x)} - \frac{1}{h_n(x)} \right| \leq o \left(\frac{n}{k_n \log n} \right) \text{ a.s.} \quad (11)$$

and

$$\sup_{x \in [T'', T]} |h_n(x) - a_n(x)| \leq o(k_n(n \log n)^{-1}) \text{ a.s.} \quad (12)$$

Therefore, using integration by parts and the conditions given for the kernel K and k_n , we can readily establish, by lemma 1 and (12), that

$$\begin{aligned} I_1 &= \sup_{x \in [T'', T]} \frac{1}{a_n(x)} \left| \int_{|s| \leq 1} K(s) d(\alpha_n(x - a_n(x)s) - \alpha_n(x - h_n(x)s)) \right| \\ &\leq \frac{cn}{k_n^{\frac{1}{2}}} \sup_{\mathcal{A}'_n} |F_n(A) - F(A)| \rightarrow 0 \text{ a.s.,} \end{aligned}$$

where $\mathcal{A}'_n = \{A : \exists x \in [T'', T] \text{ such that } A \text{ is an interval and}$

$A \subset (x - o(k_n(n \log n)^{-1}), x + o(k_n(n \log n)^{-1}))\}$ and so all intervals A in $\mathcal{A}'_n$ with mass less than or equal to $o(k_n(n \log n)^{-1})$. Similar to the proof of I_1 , we have $I_2 \rightarrow 0$ a.s. by lemma 1 and (11). Hence (10) holds.

2. We show that

$$\left(\frac{k_n}{2 \log(n/k_n)} \right)^{\frac{1}{2}} \sup_{T'' \leq x \leq T} \frac{1}{f(x)} |f_n^*(x) - f_n^{\#}(x)| \rightarrow \left(2 \int K^2(x) dx \right)^{\frac{1}{2}} \text{ a.s.} \quad (13)$$

Since $f(x)$ and $\bar{G}(x)$ are continuous on the interval $[T'', T']$, then for any $\varepsilon > 0$, there is a $\eta > 0$, such that $|f(x)\bar{G}(x) - f(y)\bar{G}(y)| \leq \varepsilon$ if $|x - y| \leq \eta$. Now we divide the interval $[T'', T]$ into m sub-intervals $[t_i, t_{i+1}]$ of length at most η with $t_1 = T''$ and $t_{m+1} = T$.

Define

$$f_{ni}^*(x) = \frac{1}{h_n(t_i)} \int K\left(\frac{x-s}{h_n(t_i)}\right) dF_n(s),$$

and

$$f_n^*(x) = \frac{1}{h_n(t_i)} \int K\left(\frac{x-s}{h_n(t_i)}\right) dF(s).$$

Note that

$$\sup_{x \in [t_i, t_{i+1}]} |h_n(x) - h_n(t_i)| \leq \varepsilon M^2 k_n / 2n$$

and

$$\sup_{x \in [t_i, t_{i+1}]} |h_n^{-1}(x) - h_n^{-1}(t_i)| \leq 2\varepsilon n / k_n,$$

we can obtain that

$$\sup_{t_i \leq x \leq t_{i+1}} \left(\frac{nh_n(x)}{\log(n/k_n)} \right)^{\frac{1}{2}} \sqrt{\frac{\bar{G}(x)}{f(x)}} |(f_n^*(x) - \bar{f}_n^*(x)) - (f_{ni}^*(x) - \bar{f}_{ni}^*(x))| \leq \sqrt{\varepsilon} \quad (14)$$

for $i = 1, \dots, m$ by an analogous argument in the proof of 1. Thus, by (3) and (14), for n large enough,

$$\begin{aligned} & \left(\frac{k_n}{2 \log(n/k_n)} \right)^{\frac{1}{2}} \sup_{T'' \leq x \leq T} \frac{1}{f(x)} |f_n^*(x) - \bar{f}_n^*(x)| \\ &= \sup_{T'' \leq x \leq T} \left(\frac{k_n}{2 \log(n/k_n)} \right)^{\frac{1}{2}} \frac{1}{f(x)} |f_n^*(x) - \bar{f}_n^*(x)| \\ &= \sup_{T'' \leq x \leq T} \left(\frac{nh_n(x)}{2 \log(n/k_n)} \right)^{\frac{1}{2}} \sqrt{\frac{\bar{G}(x)}{f(x)}} |f_n^*(x) - \bar{f}_n^*(x)| \\ &= \max_{1 \leq i \leq m} \sup_{t_i \leq x \leq t_{i+1}} \left(\frac{nh_n(x)}{\log(n/k_n)} \right)^{\frac{1}{2}} \sqrt{\frac{\bar{G}(x)}{f(x)}} \\ & \quad \times |(f_n^*(x) - \bar{f}_n^*(x)) - (f_{ni}^*(x) - \bar{f}_{ni}^*(x)) + (f_{ni}^*(x) - \bar{f}_{ni}^*(x))| \\ &\leq \sqrt{\varepsilon} + \max_{1 \leq i \leq m} \sup_{t_i \leq x \leq t_{i+1}} \left(\frac{h_n(x)}{h_n(t_i)} \right)^{\frac{1}{2}} \left(\frac{nh_n(t_i)}{\log(n/k_n)} \right)^{\frac{1}{2}} \end{aligned}$$

$$\begin{aligned}
& \times \sqrt{\frac{\bar{G}(x)}{f(x)}} |f_{ni}^*(x) - \bar{f}_{ni}^*(x)| \\
& = \sqrt{\varepsilon} + ((1 + M\varepsilon))^{\frac{1}{2}} \left(2 \int K^2(x) dx \right)^{\frac{1}{2}} \text{ a.s.,} \tag{15}
\end{aligned}$$

where $M = 1/m\bar{H}(T'')$. On the other hand, similar to the argument in (15), also by (3) and (14), for n large enough,

$$\begin{aligned}
& \left(2 \int K^2(x) dx \right)^{\frac{1}{2}} \\
& = \max_{1 \leq i \leq m} \sup_{t_i \leq x \leq t_{i+1}} \left(\frac{nh_n(t_i)}{\log(n/k_n)} \right)^{\frac{1}{2}} \sqrt{\frac{\bar{G}(x)}{f(x)}} |f_{ni}^*(x) - \bar{f}_{ni}^*(x)| \\
& \leq \sqrt{\varepsilon} + \max_{1 \leq i \leq m} \sup_{t_i \leq x \leq t_{i+1}} \left(\frac{h_n(t_i)}{h_n(x)} \right)^{\frac{1}{2}} \left(\frac{nh_n(x)}{\log(n/k_n)} \right)^{\frac{1}{2}} \sqrt{\frac{\bar{G}(x)}{f(x)}} \\
& \quad \times |f_n^*(x) - \bar{f}_n^*(x)| \\
& \leq \sqrt{\varepsilon} + (1 + M\varepsilon)^{\frac{1}{2}} \max_{1 \leq i \leq m} \sup_{t_i \leq x \leq t_{i+1}} \left(\frac{nh_n(x)}{\log(n/k_n)} \right)^{\frac{1}{2}} \sqrt{\frac{\bar{G}(x)}{f(x)}} \\
& \quad \times |f_n^*(x) - \bar{f}_n^*(x)| \\
& = \sqrt{\varepsilon} + (1 + M\varepsilon)^{\frac{1}{2}} \left(\frac{k_n}{2 \log(n/k_n)} \right)^{\frac{1}{2}} \sup_{T'' \leq x \leq T} \frac{1}{f(x)} \\
& \quad \times |f_n^*(x) - \bar{f}_n^*(x)| \text{ a.s.} \tag{16}
\end{aligned}$$

Let $\varepsilon \rightarrow 0$, now (13) follows from (15) and (16). Combining (10) and (13), we conclude that (5) is true. (6) follows in an analogous way.

PROOF OF COROLLARY 1. Imitating the proof of theorem 1 and using the definition of $\bar{f}_n(x)$ and (7), we need only to consider that

$$\left(\frac{k_n}{2 \log(n/k_n)} \right)^{\frac{1}{2}} \sup_{T'' \leq x \leq T} |\bar{f}_n(x) - f(x)|$$

$$\begin{aligned}
&\leq \left(\frac{k_n}{2 \log(n/k_n)} \right)^{\frac{1}{2}} \sup_{T'' \leq x \leq T} \int_{-1}^1 K(s) |f(x - a_n(x)s) - f(x)| ds \\
&\leq k_n^{3/2}/n \log^{\frac{1}{2}}(n/k_n) \rightarrow 0.
\end{aligned} \tag{17}$$

PROOF OF THEOREM 2. Define

$$\begin{aligned}
g_n^*(x) &= \frac{1}{q_n(x)} \int K\left(\frac{x-s}{q_n(x)}\right) dF_n(s), \\
\bar{g}_n^*(x) &= \frac{1}{q_n(x)} \int K\left(\frac{x-s}{q_n(x)}\right) dF(s).
\end{aligned}$$

where $q_n(x)$ is defined by $H(x + q_n(x)) - H(x - q_n(x)) = k_n/n$. It follows from Proposition 2 of Mack (1983), for $\xi \geq 0$ sufficiently large, that

$$\sup_{x \in [T'', T]} |\rho_n(x) - q_n(x)| \leq n^{-1} [k_n(1 + \xi) \log n]^{\frac{1}{2}} \text{ a.s.}$$

and

$$\sup_{x \in [T'', T]} |\rho_n^{-1}(x) - q_n^{-1}(x)| \leq O\left(nk_n^{-3/2}((1 + \xi) \log n)^{\frac{1}{2}}\right) \text{ a.s.}$$

We conclude the proof of (8) and (9) by a similar argument to that used in the proof of theorem 1.

PROOF OF COROLLARY 2. By the definition of $q_n(x)$, we see that $q_n(x) = O(k_n/n)$ uniformly for $x \in [T'', T]$. Thus a similar argument as in (17) yields corollary 2.

References

- DIEHL, S. & STUTE, W. (1988). Kernel density estimation in the presence of censoring. *J. Multivar. Anal.*, **25**, 299–310.
- HALL, P. (1981). Laws of the iterated logarithm for nonparametric density estimators. *Z. Wahrsch. Verm. Gebiete.*, **56**, 47–61.

- KAPLAN, E.L. & MEIER, P. (1958). Nonparametric estimation from incomplete observations. *J. Am. Statist. Ass.*, **53**, 457–481.
- LOFTSGAARDEN, D.O. & QUESENBERRY, C.P. (1965). A nonparametric estimate of a multivariate density function. *Ann. Math. Statist.*, **36**, 1049–1051.
- MACK, YP. (1983). Rate of strong uniform convergence of k -NN density estimates. *J. Statist. Plann. Inference*, **8**, 185–192.
- PARZEN, E. (1962). On estimation of a probability density function and mode. *Ann. Math. Statist.*, **23**, 1065–1076.
- RALESCU, S.S. (1995). The law of the iterated logarithm for the multivariate nearest neighbor density estimators. *J. Multivar. Anal.*, **53**, 159–179.
- ROSSITER, J.E. (1991). Calculating centile curves using kernel density estimation methods with application to infant kidney lengths. *Statist. Med.*, **10**, 1693–1701.
- SCHÄFER, H. (1986). Local convergence of empirical measures in the random censorship situation with application to density and rate estimators. *Ann. Statist.*, **14**, 1240–1245.
- STUTE, W. (1982). A law of the logarithm for kernel density estimators. *Ann. Probab.*, **10**, 414–422.
- TANNER, M.A. (1983). a note on the variable kernel estimator of the hazard function from randomly censored data. *Ann. Statist.*, **11**, 994–997.
- ZHANG, B. (1996). A law of the iterated logarithm for kernel estimators of hazard functions under random censorship. *Scand. J. Statist.*, **23**, 37–47.